Spatial Estimates of SWE for the Rio Grande Basin

April 22, 2025

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Overview Map – Rio Grande Basin Study Area

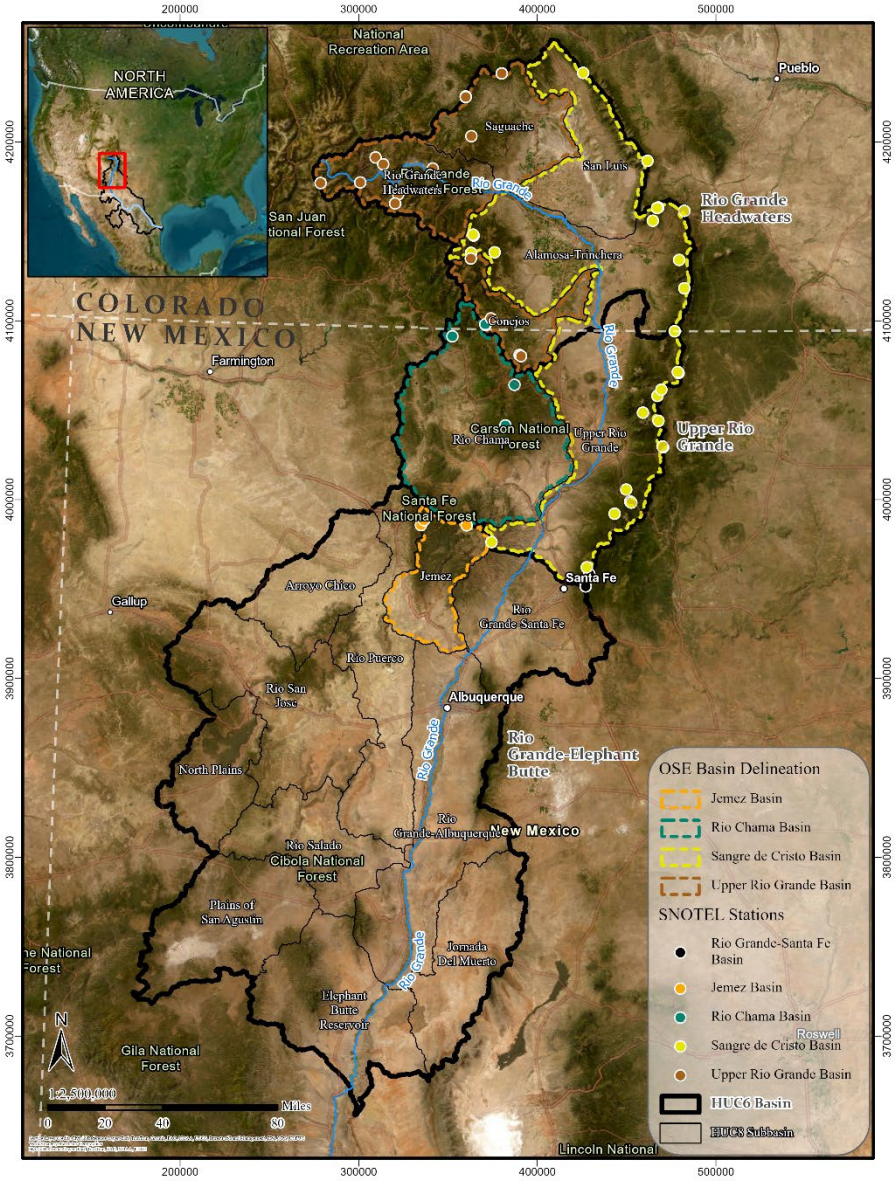

Figure 1. Overview map of the Rio Grande (RG) Basin study area. *The Rio Grande Headwaters, Upper Rio Grande, and Rio Grande-Elephant Butte Basins are presented as the standard HUC6 basins. HUC8 groupings (Jemez – orange, Rio Chama – green, Sangre de Cristo – yellow, and Upper Rio Grande – brown) are symbolized.*

Summary of SNODAS SWE for April 22, 2025

- Spatial estimates of snow water equivalent (SWE) and change in SWE (presented in inches) for the Rio Grande (RG) Basin study area are presented in Figure 2.
- A spring storm brought new snow to the region on Friday (April 18, 2025) and Saturday (April 19, 2025).
- The spring storm occurred in between periods of warm and dry weather, resulting in a decrease in estimated SNODAS SWE in the Rio Grande Headwaters Basin (HUC6) and Upper Rio Grande Basin (HUC6), with the only increase occurring in the Alamosa-Trinchera and San Luis subbasins (Figures 3 and 4; Tables 1 and 2).
- It appears that the lower elevations in the Rio Grande Headwaters Basin (HUC6) and Upper Rio Grande Basin (HUC6) gained SWE, while higher elevations lost SWE (Figures 3 and 4; Table 3). There is some additional uncertainty in the estimated SWE because of this pattern, which may be caused by new snowfall occurring in-between periods of warm weather and large-scale snowmelt.
- The storm brought enough new snow to the Rio Grande-Elephant Butte Basin (HUC6), resulting in an increase in estimated SNODAS SWE, with a majority of the SWE present in the Jemez subbasin (Figure 5; Tables 1 and 2).
- **Table 1**: Estimated SWE amount (median and mean in inches (in)), estimated volume (acre-feet (af)), and change in volume (af) since the last report, are shown for the standard Hydrologic Unit Code 6 (HUC6) basins (Rio Grande Headwaters (Figure 3), Upper Rio Grande (Figure 4), and Rio Grande-Elephant Butte (Figure 5)).
- **Table 2**: Estimated SWE amount (median and mean (in)), estimated volume (af), and change in volume (af) since the last report, are shown for the specific HUC8 subbasin groupings (OSE Basin Delineations shown in Figure 1; Upper Rio Grande, Rio Chama, Sangre de Cristo, and Jemez).
- **Table 3**: Estimated SWE amount (median and mean (in)), estimated volume (af), and change in volume (af) since the last report, are shown by basin and elevation band for the HUC6 basins (Rio Grande Headwaters (Figure 3), Upper Rio Grande (Figure 4), and Rio Grande-Elephant Butte (Figure 5)).
- **Rio Grande Headwaters (HUC6)**: SNODAS SWE volume for April 22, 2025, is estimated to be 495,015 af, a decrease of an estimated 33,438 af from April 15, 2025.
- **Upper Rio Grande (HUC6)**: SNODAS SWE volume for April 22, 2025, is estimated to be 120,484 af, a decrease of an estimated 12,966 af from April 15, 2025.
- **Rio Grande-Elephant Butte (HUC6)**: SNODAS SWE volume for April 22, 2025, is estimated to be 5,219 af, an increase of an estimated 5,088 af from April 15, 2025.

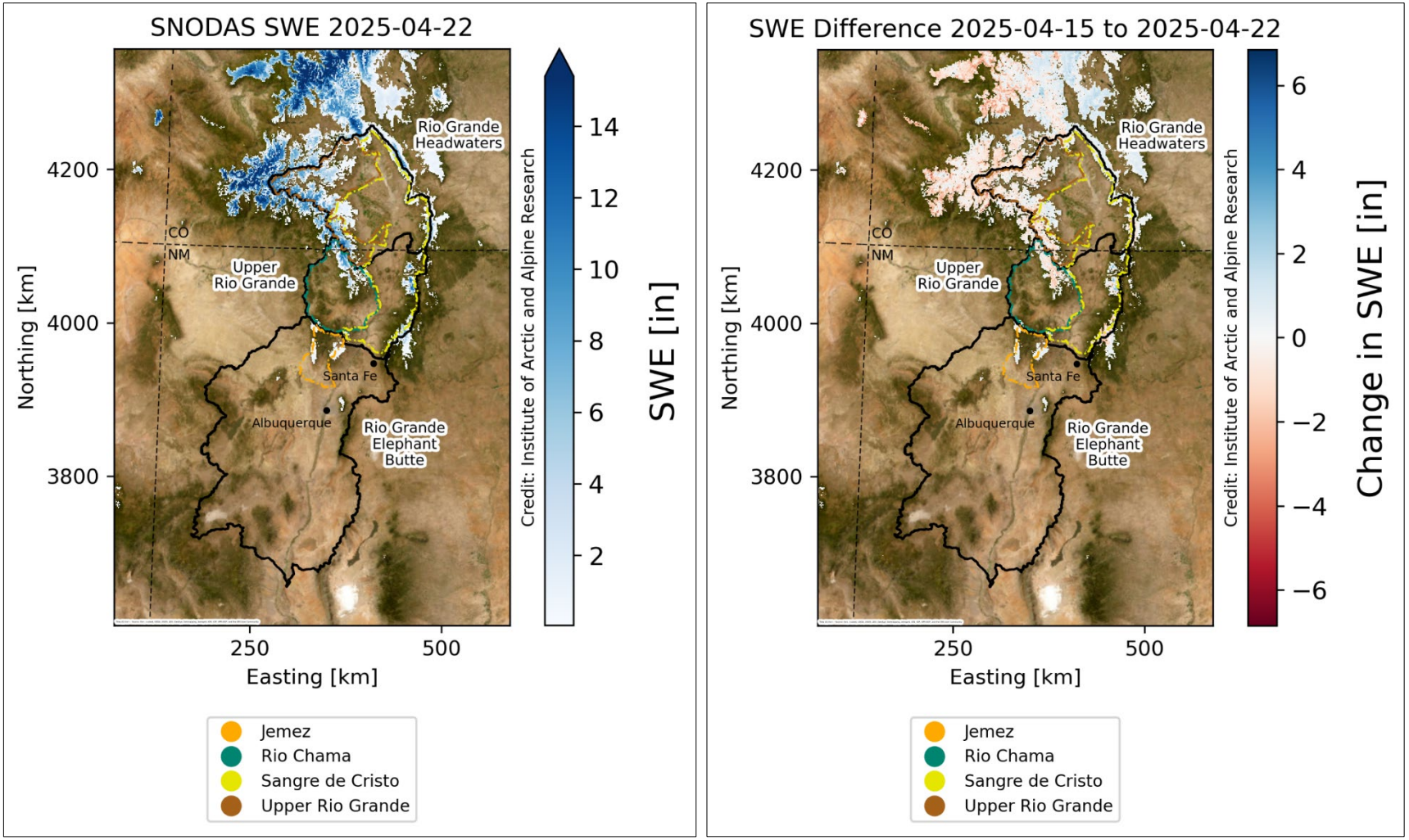

Figure 2. Estimated SNODAS SWE and Change in SNODAS SWE across the Rio Grande Basin study area. *SWE amounts (in) across the RG Basin study area (left); dark blue pixels denote greater SWE amounts. Change in SNODAS SWE amounts (in) across the RG Basin study area from April 15, 2025, to April 22, 2025 (right); blue pixels denote an increase in SWE amounts whereas red pixels denote a decrease in SWE amounts.*

Introduction

The Rio Grande (RG) is a snowmelt-driven river system with up to 75% of its water originating from seasonal snowpack in the mountains of Colorado and northern New Mexico (Moeser et al., 2021; Figure 2). Snowpack acts as a natural reservoir in the RG Basin, gradually releasing meltwater during the spring and early summer months, sustaining streamflow and providing vital water supply for downstream users. However, regional decreases in snowpack volume and earlier snowmelt timing, especially throughout the western U.S. and New Mexico, can lead to reduced water availability (Mote et al., 2005; EPA, 2016b; Mote et al., 2018; Hale et al., 2023). Long-term changes in snowpack patterns and timing can serve as an indicator of climate change, while understanding short-term changes is critical for monitoring seasonal variability and forecasting seasonal water supply. Monitoring snowpack provides valuable data on shifts in precipitation patterns and temperatures, helping scientists and decision-makers better understand and manage climate-related challenges.

Snow water equivalent (SWE) is the estimate of the depth of liquid water contained in the snowpack. SWE is the standard metric in estimating the volume of meltwater in the snowpack, which plays a critical role in the seasonal snow water supply in the RG Basin. SWE is measured at in-situ Snow Telemetry (SNOTEL) sites throughout the RG Basin and the greater Western U.S. However, the spatial distribution of SWE is complex and often difficult to estimate outside of the in-situ SNOTEL measurements. While SNOTEL provides important point-based measurements, it has been suggested that these data may not be representative of the surrounding snow-covered area (Meromy et al., 2013). Whereas regional tendencies of spatial distribution are consistent, local snowpack volumes are more variable, all of which influences the magnitude and timing of snowmelt-driven runoff.

SWE Analysis Tool

The RG Basin, and more broadly the Western U.S., is experiencing a prolonged and severe drought (EPA, 2016a), exacerbated by a warming climate that increases evapotranspiration, reduces overall snowpack quantity, and reduces the seasonal duration of snowpack. The application of the SWE Analysis Tool (SWEAT) will supply scientists and decision-makers with an improved insight into the spatial distribution of SWE by providing real-time spatial SWE estimates for the RG Basin, its subbasins, and elevation bands within the basin.

Currently, SWEAT incorporates the SNODAS gridded SWE data product (NOHRSC, 2004). Because no spaceborne sensor can directly measure SWE, spatial distribution of SWE must be estimated from remotely sensed observations coupled with modeling approaches. However, gridded SWE products, like SNODAS, are known to have accuracy issues across both mountainous and non-mountainous regions (Mudryk et al., 2025), resulting in a range of spatial SWE estimates for any given location. NM OSE and INSTAAR are working to integrate more data products to build SWEAT based on an ensemble of spatially estimated SWE. By taking an ensemble approach, the goal for SWEAT is to provide spatial SWE estimations for the RG Basin that are, on average, equally or more accurate than any individual model (Thompson, 1977; Arsenault et al., 2015). In addition, the use of a single estimate calculated from an ensemble can reduce confusion for water managers about which product may be performing best, providing a path towards consistency throughout users, and is useful in identifying product outliers.

SNODAS

The SNOw Data Assimilation System (SNODAS) represents a modeling and data assimilation system developed by the National Operational Hydrologic Remote Sensing Center (NOHRSC). SNODAS generates spatial estimates of SWE, snow cover, and snow depth through a combination of remotely sensed snow data and in-situ measurements of SWE from SNOTEL at a 1-km spatial resolution and 24-hour temporal resolution.

The national SNODAS dataset is available through the SNODAS Data Products at National Snow and Ice Data Center (NSIDC), Version 1 (NOHRSC, 2004). SNODAS data can be viewed at the NOHRSC Interactive Snow Information page. Gridded observed snowfall images, seasonal totals, and data downloads in several formats can be found on the National Gridded Snowfall Analysis page.

Refer to the SNODAS User Guide and the NSIDC Special Report 11 (Barrett, 2003) for more information.

Discussion

Figures and tables for the identified basins (Figure 1) are presented below. The median and mean SWE (in) is presented to avoid potentially skewed outliers, most notably due to an uneven distribution of SWE across elevation bands. While the mean value represents the average value, it is highly influenced by outliers and is typically best used for uniform distributions of data. In contrast, the median value represents the middle value and is less sensitive to outliers, which can be more useful in skewed distributions. Note that the basin-wide median and mean values may not be directly comparable with the SNOTEL basin-wide values. If comparisons are made with SNOTEL sites, a better comparison may be made across elevation bands (Table 3).

Figure 3 (Rio Grande Headwaters), Figure 4 (Upper Rio Grande), and Figure 5 (Rio Grande-Elephant Butte) show the spatial distribution of SWE amount (in), change in SWE volume (af) since the last report, and SWE volume (af) across subbasin and elevation. Since the last report, a spring storm brought new snow to the RG Basin. However, the spring storm occurred in between periods of warm and dry weather, resulting in a decrease in estimated SNODAS SWE in the Rio Grande Headwaters Basin (HUC6) and Upper Rio Grande Basin (HUC6), with the only increase in estimated SNODAS SWE occurring in the Alamosa-Trinchera and San Luis subbasins (Figures 3 and 4; Tables 1 and 2). It appears that the lower elevations in the Rio Grande Headwaters Basin (HUC6) and Upper Rio Grande Basin (HUC6) gained SWE, while higher elevations lost SWE (Figures 3 and 4; Table 3). There is some additional uncertainty in the estimated SWE because of this pattern. Despite the warm weather patterns, the storm brought enough new snow to the Rio Grande-Elephant Butte Basin (HUC6), resulting in an increase in estimated SNODAS SWE, with a majority of the SWE present in the Jemez subbasin (Figure 5; Tables 1 and 2).

Estimated SWE amounts (in), volumes (af), and change in volumes (af) since the last report are presented in the tables below.

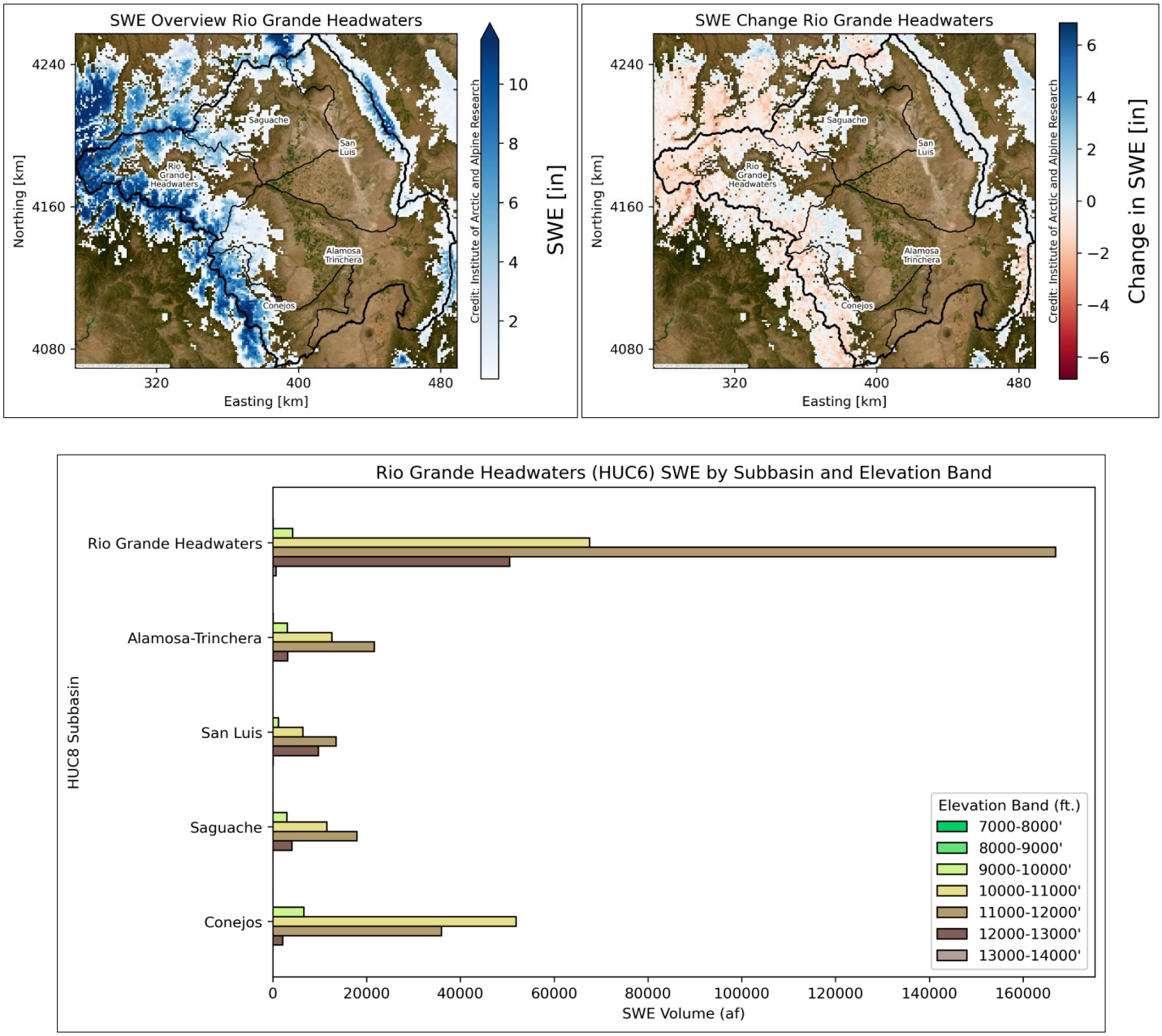

Figure 1. Estimated SNODAS SWE (in) and SWE volume (af) across the Rio Grande Headwaters Basin for April 22, 2025. *SWE amount (top left) and change in SWE since the last report (top right) across the Rio Grande Headwaters Basin (HUC6), including the Rio Grande Headwaters, Alamosa-Trinchera, San Luis, Saguache, and Conejos subbasins. SWE volume (af) by subbasin and elevation band across the Rio Grande Headwaters Basin (bottom).*

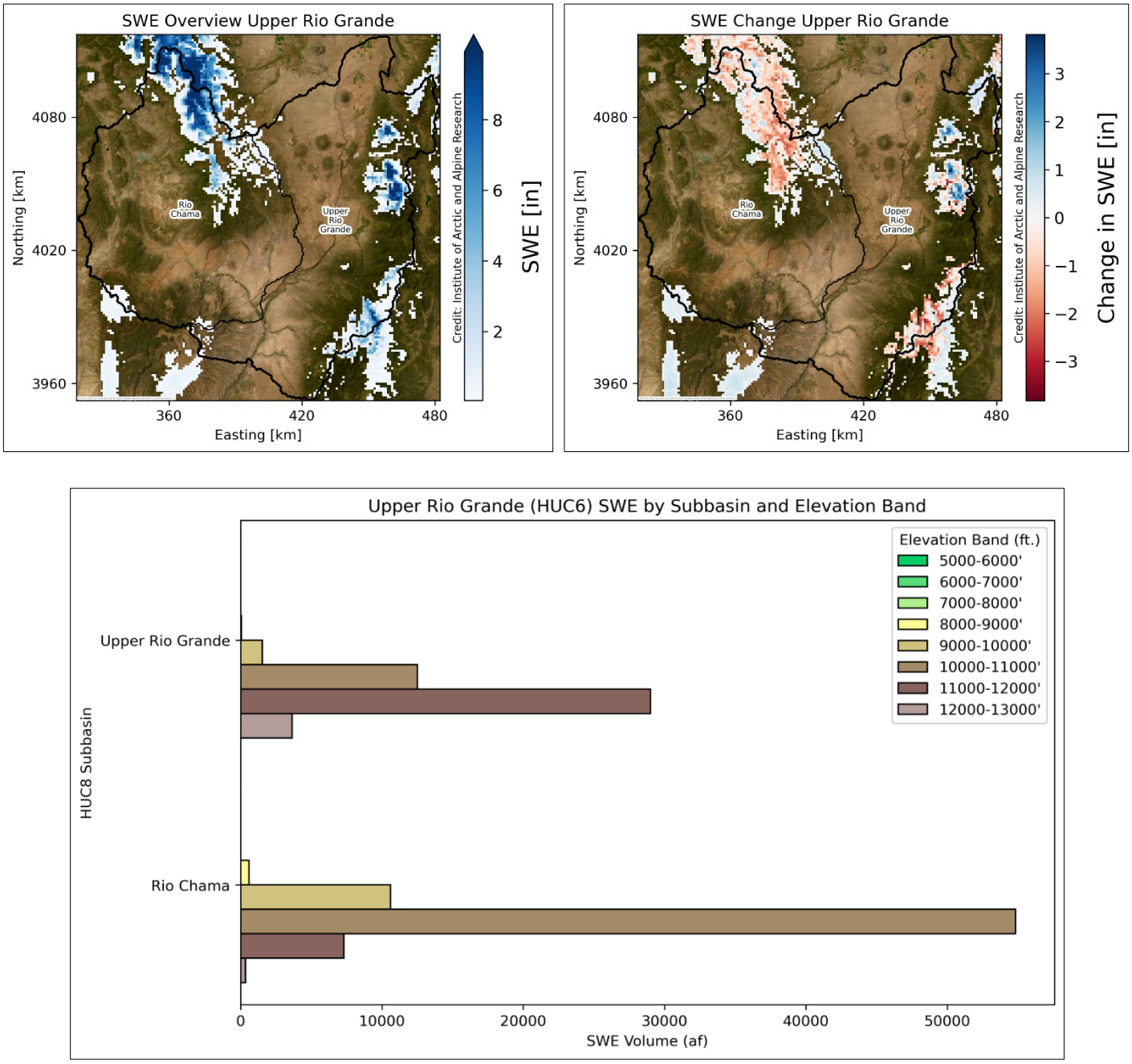

Figure 2. Estimated SNODAS SWE (in) and SWE volume (af) across the Upper Rio Grande Basin for April 22, 2025. *SWE amount (top left) and change in SWE since the last report (top right) across the Upper Rio Grande Basin (HUC6), including the Upper Rio Grande and Rio Chama subbasins. SWE volume (af) by subbasin and elevation band across the Upper Rio Grande Basin (bottom).*

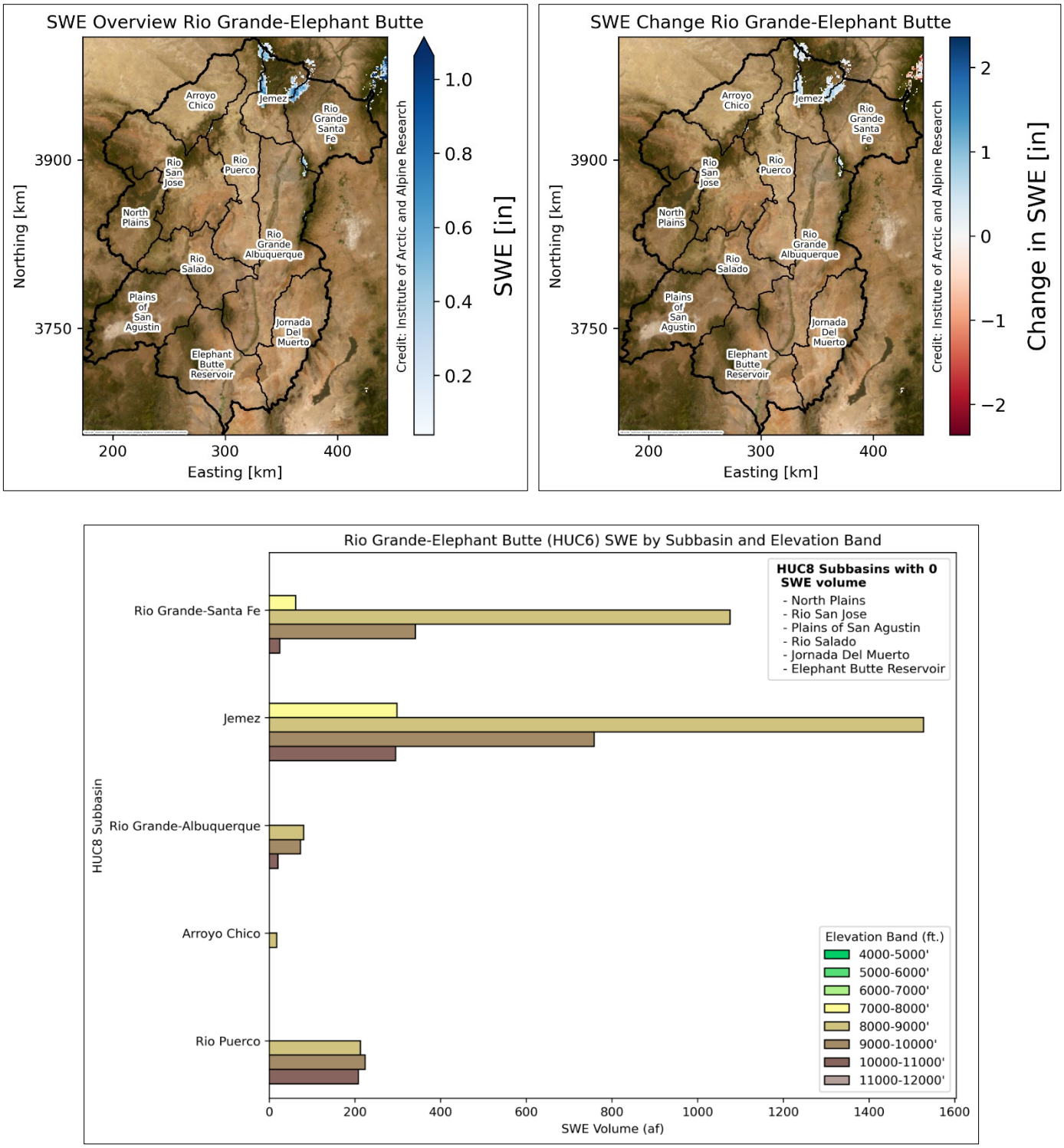

Figure 3. Estimated SNODAS SWE (in) and SWE volume (af) across the Rio Grande-Elephant Butte Basin for April 22, 2025. *SWE amount (top left) and change in SWE since the last report (top right) across Rio Grande-Elephant Butte (HUC6), including the Rio Grande-Santa Fe, Jemez, Rio Grande-Albuquerque, Arroyo Chico, North Plains, Rio San Jose, Plains of San Agustin, Rio Salado, Jornada Del Muerto, Elephant Butte Reservoir, and Rio Puerco subbasins. SWE volume (af) by subbasin and elevation band across the Rio Grande-Elephant Butte Basin (bottom).*

Tables

Table 1. Estimated SWE by HUC6 Basin for April 22, 2025. *Shown are median SWE (in.), mean SWE (in.), estimated volume (af), and volume change (af) since the last report. Values are separated by HUC8 subbasin and totaled for each HUC8 basin (bold).*

HUC6 Basin	HUC8 Subbasin	Median SWE (in.)	Mean SWE (in.)	Estimated Volume (af)	Volume Change since April 15, 2025 (af)
Rio Grande Headwaters	Rio Grande Headwaters	2.2	3.9	290,040	-24,546
	Alamosa-Trinchera	0.0	0.3	40,702	931
	San Luis	0.0	0.4	30,949	3,815
	Saguache	0.0	0.5	36,726	-3,108
	Conejos	0.0	2.4	96,598	-10,531
	Total	**0.0**	**1.2**	**495,015**	**-33,438**
Upper Rio Grande	Upper Rio Grande	0.0	0.3	46,767	-1,290
	Rio Chama	0.0	0.4	73,716	-11,677
	Total	**0.0**	**0.4**	**120,484**	**-12,966**
Rio Grande-Elephant Butte	Rio Grande-Santa Fe	0.0	0.0	1,503	1,478
	Jemez	0.0	0.1	2,880	2,775
	Rio Grande-Albuquerque	0.0	0.0	173	173
	Arroyo Chico	0.0	0.0	18	18
	North Plains	0.0	0.0	0	0
	Rio San Jose	0.0	0.0	0	0
	Plains of San Agustin	0.0	0.0	0	0
	Rio Salado	0.0	0.0	0	0
	Jornada Del Muerto	0.0	0.0	0	0
	Elephant Butte Reservoir	0.0	0.0	0	0
	Rio Puerco	0.0	0.0	645	644
	Total	**0.0**	**0.0**	**5,219**	**5,088**

Table 2. Estimated SWE by Specified Subbasin Groupings for April 22, 2025. *Shown are median SWE (in.), mean SWE (in.), estimated volume (af), and volume change (af) since the last report. Values are separated by HUC8 subbasin and totaled for each specified subbasin grouping (bold).*

Subbasin Grouping	HUC8 Subbasin	Median SWE (in.)	Mean SWE (in.)	Estimated Volume (af)	Volume Change since April 15, 2025 (af)
Upper Rio Grande	Rio Grande Headwaters	2.2	3.9	290,040	-24,546
	Saguache	0.0	0.5	36,726	-3,108
	Conejos	0.0	2.4	96,598	-10,531
	Total	**0.1**	**2.3**	**423,364**	**-38,185**
Sangre de Cristo	San Luis	0.0	0.4	30,949	3,815
	Alamosa-Trinchera	0.0	0.3	40,702	931
	Upper Rio Grande	0.0	0.3	46,767	-1,290
	Total	**0.0**	**0.3**	**118,418**	**3,457**
Rio Chama		**0.0**	**0.4**	**73,716**	**-11,677**
Jemez		**0.0**	**0.1**	**2,880**	**2,775**

Table 3. Estimated SWE by HUC6 Basin and Elevation Band for April 22, 2025. *Shown are median SWE (in.), mean SWE (in.), estimated volume (af), and volume change (af) since the last report. Values are separated by elevation band (1000 ft. increments) and HUC8 subbasin.*

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Median SWE (in.)	Mean SWE (in.)	Estimated Volume (af)	Volume Change since April 15, 2025 (af)
Rio Grande Headwaters	Rio Grande Headwaters	7000-8000'	0.0	0.0	0	0
		8000-9000'	0.0	0.0	84	84
		9000-10000'	0.0	0.3	4,211	1,147
		10000-11000'	1.5	3.1	67,552	-11,591
		11000-12000'	7.9	7.6	166,961	-10,527
		12000-13000'	9.0	8.4	50,538	-3,617
		13000-14000'	7.5	5.6	693	-43
	Alamosa-Trinchera	7000-8000'	0.0	0.0	0	0
		8000-9000'	0.0	0.0	118	118
		9000-10000'	0.0	0.2	3,133	2,507
		10000-11000'	0.6	1.0	12,589	135
		11000-12000'	1.5	2.5	21,617	-2,215
		12000-13000'	1.0	1.7	3,197	340
		13000-14000'	0.6	0.6	48	48
	San Luis	7000-8000'	0.0	0.0	0	0
		8000-9000'	0.0	0.0	15	15
		9000-10000'	0.0	0.2	1,200	986
		10000-11000'	0.4	1.1	6,391	94
		11000-12000'	1.9	2.9	13,506	817
		12000-13000'	2.8	3.8	9,750	1,852
		13000-14000'	0.7	1.4	87	50
	Saguache	7000-8000'	0.0	0.0	0	0
		8000-9000'	0.0	0.0	5	5
		9000-10000'	0.0	0.2	3,026	2,114
		10000-11000'	0.4	0.9	11,569	-1,608
		11000-12000'	4.0	3.9	17,994	-3,207
		12000-13000'	4.4	4.4	4,121	-422
		13000-14000'	0.6	0.6	13	11
	Conejos	7000-8000'	0.0	0.0	0	0
		8000-9000'	0.0	0.0	6	-29
		9000-10000'	0.0	0.6	6,617	-1,535
		10000-11000'	5.4	5.5	51,864	-7,006

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Median SWE (in.)	Mean SWE (in.)	Estimated Volume (af)	Volume Change since April 15, 2025 (af)
	Conejos	11000-12000'	7.6	7.3	35,964	-1,812
		12000-13000'	5.4	5.5	2,147	-149
Upper Rio Grande	Upper Rio Grande	5000-6000'	0.0	0.0	0	0
		6000-7000'	0.0	0.0	0	0
		7000-8000'	0.0	0.0	0	0
		8000-9000'	0.0	0.0	66	66
		9000-10000'	0.0	0.1	1,544	548
		10000-11000'	0.0	0.7	12,502	-4,750
		11000-12000'	2.2	3.7	28,997	1,957
		12000-13000'	1.9	3.1	3,658	889
	Rio Chama	5000-6000'	0.0	0.0	0	0
		6000-7000'	0.0	0.0	0	0
		7000-8000'	0.0	0.0	0	0
		8000-9000'	0.0	0.0	614	502
		9000-10000'	0.0	0.5	10,596	-2,745
		10000-11000'	4.6	4.9	54,829	-9,467
		11000-12000'	9.5	9.1	7,313	24
		12000-13000'	8.9	8.9	366	9
Rio Grande-Elephant Butte	Rio Grande-Santa Fe	5000-6000'	0.0	0.0	0	0
		6000-7000'	0.0	0.0	0	0
		7000-8000'	0.0	0.0	62	62
		8000-9000'	0.1	0.2	1,076	1,076
		9000-10000'	0.3	0.3	341	341
		10000-11000'	0.0	0.1	24	-1
		11000-12000'	0.0	0.0	0	0
	Jemez	5000-6000'	0.0	0.0	0	0
		6000-7000'	0.0	0.0	0	0
		7000-8000'	0.0	0.0	298	298
		8000-9000'	0.0	0.1	1,527	1,527
		9000-10000'	0.0	0.1	759	722
		10000-11000'	0.4	0.3	295	228
	Rio Grande-Albuquerque	4000-5000'	0.0	0.0	0	0
		5000-6000'	0.0	0.0	0	0
		6000-7000'	0.0	0.0	0	0
		7000-8000'	0.0	0.0	0	0

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Median SWE (in.)	Mean SWE (in.)	Estimated Volume (af)	Volume Change since April 15, 2025 (af)
Rio Grande-Elephant Butte	Rio Grande-Albuquerque	8000-9000'	0.0	0.0	80	80
		9000-10000'	0.0	0.1	73	73
		10000-11000'	0.2	0.2	20	20
	Arroyo Chico	5000-6000'	0.0	0.0	0	0
		6000-7000'	0.0	0.0	0	0
		7000-8000'	0.0	0.0	0	0
		8000-9000'	0.0	0.0	18	18
		9000-10000'	0.0	0.0	0	0
		10000-11000'	0.0	0.0	0	0
	North Plains	6000-7000'	0.0	0.0	0	0
		7000-8000'	0.0	0.0	0	0
		8000-9000'	0.0	0.0	0	0
	Rio San Jose	5000-6000'	0.0	0.0	0	0
		6000-7000'	0.0	0.0	0	0
		7000-8000'	0.0	0.0	0	0
		8000-9000'	0.0	0.0	0	0
		9000-10000'	0.0	0.0	0	0
		10000-11000'	0.0	0.0	0	0
	Plains of San Agustin	6000-7000'	0.0	0.0	0	0
		7000-8000'	0.0	0.0	0	0
		8000-9000'	0.0	0.0	0	0
		9000-10000'	0.0	0.0	0	0
	Rio Salado	4000-5000'	0.0	0.0	0	0
		5000-6000'	0.0	0.0	0	0
		6000-7000'	0.0	0.0	0	0
		7000-8000'	0.0	0.0	0	0
		8000-9000'	0.0	0.0	0	0
		9000-10000'	0.0	0.0	0	0
	Jornada Del Muerto	4000-5000'	0.0	0.0	0	0
		5000-6000'	0.0	0.0	0	0
		6000-7000'	0.0	0.0	0	0
		7000-8000'	0.0	0.0	0	0
		8000-9000'	0.0	0.0	0	0
	Elephant Butte Reservoir	4000-5000'	0.0	0.0	0	0

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Median SWE (in.)	Mean SWE (in.)	Estimated Volume (af)	Volume Change since April 15, 2025 (af)
Rio Grande-Elephant Butte	Elephant Butte Reservoir	5000-6000'	0.0	0.0	0	0
		6000-7000'	0.0	0.0	0	0
		7000-8000'	0.0	0.0	0	0
		8000-9000'	0.0	0.0	0	0
		9000-10000'	0.0	0.0	0	0
		10000-11000'	0.0	0.0	0	0
	Rio Puerco	4000-5000'	0.0	0.0	0	0
		5000-6000'	0.0	0.0	0	0
		6000-7000'	0.0	0.0	0	0
		7000-8000'	0.0	0.0	0	0
		8000-9000'	0.0	0.0	213	213
		9000-10000'	0.2	0.2	224	224
		10000-11000'	0.4	0.4	208	207

Acknowledgments

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